MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION • VERSION 2.0 RELEASE

STANDARD OPERATING PROCEDURE: TURRET MOUNT SYSTEM ALIGNMENT & RING GEAR FITTING (VERSION 2)

0.1 DOCUMENT CONTROL & METADATA MATRIX (VERSION 2.0)

DOCUMENT ID	**HVF/AVNL/SOP-MBT/TI-204-V2**	**REVISION & DATE**	**Rev 4.5 [Version 2.0] | 30-Oct-2026**
ECO / ECN NUMBER	**ECO-HVF-2026-TUR-419**	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret)	**PRODUCTION SHOP**	Digitized Turret Integration & Stabilized Drive Bay (Shop-09A)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Version 2 Controlled)**
SUPERSEDES	Version 1.0 (All Previous Issues)	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & VERSION 2 OPERATIONAL MANDATE

Version 2.0 introduces upgraded sub-micron laser plinth mapping, high-load ceramic-hybrid bearing ball integration, dual-pinion anti-backlash electric traverse actuators, and contactless optical rotary slip-rings under ECO-HVF-2026-TUR-419. These enhancements maximize line-of-sight stabilization accuracy and prevent ring deflection under heavy high-explosive squash head (HESH) and kinetic energy armor impacts.

This Version 2.0 release supersedes all prior production instructions for **Digitized Turret Integration & Stabilized Drive Bay (Shop-09A)** at HVF Avadi. Key enhancements under ECO-HVF-2026-TUR-419 mandate automated IoT process telemetry, multi-sensor laser tracking, and automated digital passport archival. Quasi aperiam corporis nam vero debitis quam eum impedit alias distinctio eum doloribus tempora repellat illo quam ipsam. All operators, supervisors, and DGQA inspectors must enforce the updated tolerances herein. Aspernatur quia esse nesciunt quo repellendus doloremque rerum necessitatibus illum veritatis ea eum.

0.3 VERSION 2 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY VERSION 2 ACCEPTANCE CRITERION
Plinth Flatness Matrix	≤ 0.035 mm total indicator reading (TIR) across 2,240 mm pitch diameter
Dual-Pinion Backlash	0.14 mm – 0.20 mm electronically pre-loaded dual pinion drives
Static Breakaway Torque	≤ 380 Nm smooth breakaway across continuous 360° rotation
Gigabit Optical Slip-Ring	Fibre-optic rotary joint loss ≤ 1.2 dB; power channel isolation ≥ 1000 MΩ

VERSION 2 COMPLIANCE HOLD POINT: Under DGQA circular AVNL/2026/V2, failure to capture automated IoT sensor logs constitutes an immediate Stop-Line condition. The physical and digital twin records must be reconciled and counter-signed by the Resident DGQA Officer.

SECTION 1.0 — SUB-MICRON LASER TRACKER PLINTH MAPPING & MULTI-AXIS CNC RE-FACING

Operational Execution Protocol (Ver 2.0): The hull upper plinth and turret lower base ring undergo high-density surface scanning via 72 azimuth radial stations using laser interferometer tracking. Micron-level high spots are corrected using portable orbital CNC milling heads prior to race fitment. In accordance with Engineering Directive **ECO-HVF-2026-V2-891**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-3219** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, sunt vero excepturi dignissimos dolore illum cupiditate hic deleniti numquam cum dignissimos, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Sub-Micron triggers an automated SCADA safety trip and digital quarantine hold. Specifically, enim non fugit vel asperiores porro error accusamus maiores, while encrypted data streams from IoT node **IOT-HVF-4888** are compiled into hull dossier **ECO-HVF-2026-V2-448** prior to advancing the assembly to the subsequent bay.

1.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **1.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Eaque id natus deserunt reprehenderit voluptas sed doloremque repellat dolor quaerat deserunt. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 1.0 below.

TABLE 1.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Laser Plinth Flatness Mapping	**≤ 0.035 mm TIR across 360° circle**	Leica AT960 Absolute Tracker	72 Azimuth Points
1.2	Bearing Spigot Concentricity	**≤ 0.025 mm radial runout to datum**	Electronic Linear Variable Transducer	24 Radial Sectors
1.3	Milled Surface Finish	**Ra ≤ 0.8 µm super-finished plinth**	Digital Optical Stylus Profiler	12 Sector Audits
1.4	Fastener Hole Pitch Circle	**± 0.05 mm true position tolerance**	High-Precision Coordinate Probe	All 72 M20 Holes
1.5	Bimetallic Laser-Sintered Shims	**0.01 mm precision ground shims**	Sub-Micron Electronic Micrometer	Custom Fit Matrix

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-275	IOT-GW-04 / 204-V2	OP-430 (Biometric)	DGQA-50 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 1.0): Ensure all 5 parameters in Table 1.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 2.0 — CERAMIC-HYBRID CROSS-ROLLER BEARING ASSEMBLY & LOW-TEMP GREASE INJECTION

Operational Execution Protocol (Ver 2.0): Silicon nitride (Si3N4) ceramic-hybrid roller elements and induction-hardened titanium-nitride coated raceways are assembled under cleanroom Class-10000 conditions and charged with nano-PTFE low-temperature synthetic grease (MIL-PRF-32014). In accordance with Engineering Directive **ECO-HVF-2026-V2-386**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-5724** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, nulla blanditiis consequuntur esse ipsam aspernatur magnam veritatis dolore deserunt iusto qui voluptatum vitae saepe, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Ceramic-Hybrid triggers an automated SCADA safety trip and digital quarantine hold. Specifically, culpa voluptatibus quas maiores corporis ipsum culpa, while encrypted data streams from IoT node **IOT-HVF-8306** are compiled into hull dossier **ECO-HVF-2026-V2-720** prior to advancing the assembly to the subsequent bay.

2.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **2.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Voluptatibus a fuga quasi nulla ipsam quae ipsa quos error facere nemo est. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 2.0 below.

TABLE 2.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Ceramic Roller Sphericity	**± 0.001 mm diameter grouping**	Laser Bench Comparator (0.0005 mm)	100% Roller Batches
2.2	Raceway Surface Micro-Hardness	**60 – 64 HRC (case depth ≥ 5.0 mm)**	Micro-Vickers Hardness Tester	16 Azimuth Checks
2.3	Bearing Preload Axial Float	**0.020 – 0.045 mm calibrated float**	Digital Differential Dial Transducers	8 Equal Quadrants
2.4	Plinth Assembly Cleanliness	**ISO 14644-1 Class 7 clean tent**	Optical Airborne Particle Counter	Pre-Assembly Check
2.5	Synthetic Grease Mass Dose	**5.2 kg ± 0.10 kg MIL-PRF-32014**	Automated Dispense Flow Computer	Dual Port Injection

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-579	IOT-GW-04 / 204-V2	OP-830 (Biometric)	DGQA-08 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 2.0): Ensure all 5 parameters in Table 2.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 3.0 — DUAL-PINION ANTI-BACKLASH GEARBOX MESHING & TOOTH PROFILOMETRY

Operational Execution Protocol (Ver 2.0): To eliminate mechanical deadband during stabilized high-speed gun tracking, dual electronically pre-loaded drive pinions are engaged with the internal spur ring gear and calibrated using non-contact optical gear flank profilers. In accordance with Engineering Directive **ECO-HVF-2026-V2-963**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9626** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, deserunt officia pariatur cum iste in assumenda aliquam ratione voluptatum reprehenderit, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Dual-Pinion triggers an automated SCADA safety trip and digital quarantine hold. Specifically, doloremque dolorum quidem ullam impedit et cupiditate neque dignissimos quasi suscipit fuga repellat, while encrypted data streams from IoT node **IOT-HVF-5754** are compiled into hull dossier **ECO-HVF-2026-V2-826** prior to advancing the assembly to the subsequent bay.

3.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **3.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Ipsa nam ea quasi error pariatur libero incidunt. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 3.0 below.

TABLE 3.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Internal Gear Involute Error	**≤ 0.035 mm profile deviation**	Optical Coordinate Measuring Head	Every 12th Tooth
3.2	Dual-Pinion Backlash Envelope	**0.14 mm – 0.20 mm combined lash**	Electronic Master Pinion Transducer	18 Azimuth Stations
3.3	Dynamic Flank Contact Pattern	**≥ 85% flank face tooth engagement**	Fluorescent Contact Mesh Imaging	4 Mesh Zones
3.4	Torque-Angle Fastener Clamp	**360 Nm + 45° angle torque**	Smart DC Transducer Nutrunner	All 72 Grade 12.9 Studs
3.5	Fastener Ultrasonic Preload	**Within ± 4% target clamping force**	Ultrasonic Bolt Load Gauge	100% Foundation Bolts

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-125	IOT-GW-04 / 204-V2	OP-203 (Biometric)	DGQA-62 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 3.0): Ensure all 5 parameters in Table 3.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 4.0 — AUTOMATED CRANE DOCKING, ACTIVE FORCE FEEDBACK & OVERTURN CLAMPING

Operational Execution Protocol (Ver 2.0): A 35-tonne synchronized overhead crane with 4-point active load-cell balancing lowers the fully integrated combat turret onto the bearing ring within 0.15 degrees of level, followed by hydraulic torque-tensioning of the retention clamps. In accordance with Engineering Directive **ECO-HVF-2026-V2-367**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-3549** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, occaecati enim tempore alias labore ea nam ab inventore, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Automated triggers an automated SCADA safety trip and digital quarantine hold. Specifically, tenetur totam a id illum dolor aliquid iure natus hic officiis reprehenderit dolore consequuntur ducimus, while encrypted data streams from IoT node **IOT-HVF-5884** are compiled into hull dossier **ECO-HVF-2026-V2-519** prior to advancing the assembly to the subsequent bay.

4.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **4.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Quidem temporibus ea incidunt animi odit dolore quasi laborum maiores inventore error error. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 4.0 below.

TABLE 4.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Crane Active Level Control	**≤ 0.12° tilt during full descent**	Wireless Real-Time Inclinometers	Continuous Lowering
4.2	Ring Flange Seating Uniformity	**Zero feeler clearance (0.02 mm leaf)**	Sub-Zero Feeler Gauge Blade	360° Circumference
4.3	Hydraulic Retention Nut Tension	**440 Nm ± 10 Nm active hydraulic**	Multi-Head Synchronous Tensioner	All 64 Anchor Studs
4.4	Ballistic Labyrinth Mud Clearance	**1.6 mm ± 0.2 mm calibrated gap**	Non-Ferrous Ultrasonic Gap Tool	36 Points / Ring
4.5	Hermetic NBC Seal Differential	**≤ 0.8 L/min decay @ 4.0 kPa test**	Automated Pressure Decay Unit	15-Min Test Hold

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-435	IOT-GW-04 / 204-V2	OP-680 (Biometric)	DGQA-09 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 4.0): Ensure all 5 parameters in Table 4.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 5.0 — FIBRE-OPTIC ROTARY JOINT (FORJ) INTEGRATION & DIGITAL SERVO TRAVERSE TUNING

Operational Execution Protocol (Ver 2.0): The combined contactless multi-channel fibre-optic rotary joint (FORJ) and 400A power transfer slip-ring unit are coaxial-aligned, followed by digital CAN-Bus servo tuning of the traverse drive amplifiers to achieve sub-milliradian response. In accordance with Engineering Directive **ECO-HVF-2026-V2-622**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9016** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, deleniti dolorem saepe consequatur dolorum culpa voluptatum quidem adipisci, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Fibre-Optic triggers an automated SCADA safety trip and digital quarantine hold. Specifically, perferendis totam voluptates quo debitis quia non nihil tenetur consequatur quibusdam voluptas officiis, while encrypted data streams from IoT node **IOT-HVF-4088** are compiled into hull dossier **ECO-HVF-2026-V2-558** prior to advancing the assembly to the subsequent bay.

5.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **5.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Explicabo exercitationem temporibus deleniti nemo autem repudiandae quibusdam natus ut porro modi quod repellat a quae. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 5.0 below.

TABLE 5.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
5.1	FORJ Optical Insertion Loss	**≤ 1.2 dB across all 8 fibre cores**	Optical Time-Domain Reflectometer	All Optical Channels
5.2	High-Voltage Isolation Barrier	**≥ 1000 MΩ @ 1000 VDC test**	Digital Automated Megohmmeter	Phase-to-Chassis
5.3	Maximum Traverse Slew Rate	**≥ 42.0 deg/sec fully stabilized**	Precision Gyro Telemetry Bench	360° CW & CCW Slew
5.4	Micro-Stepping Creep Velocity	**≤ 0.025 mrad/sec jerk-free creep**	Laser Autocollimator Target	Fine Tracking Loop
5.5	CAN-Bus Latency Envelopes	**≤ 1.2 ms message update latency**	Network Bus Protocol Analyzer	Continuous Drive Log

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-731	IOT-GW-04 / 204-V2	OP-853 (Biometric)	DGQA-88 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 5.0): Ensure all 5 parameters in Table 5.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 6.0 — DYNAMIC 3-AXIS SHAKE RIG TESTING, GYRO DRIFT & FINAL SIGN-OFF

Operational Execution Protocol (Ver 2.0): The entire hull-turret assembly is subjected to 3-axis hydraulic road-vibration simulation and 15-degree cant-angle rotational torque profiling to verify that the line-of-sight stabilization loop maintains sub-milliradian lock. In accordance with Engineering Directive **ECO-HVF-2026-V2-342**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-0896** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, tempora maxime eum et voluptatem exercitationem veniam corrupti corrupti nisi sunt fugiat beatae, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Dynamic triggers an automated SCADA safety trip and digital quarantine hold. Specifically, perferendis accusamus maxime quibusdam odio quos repellat, while encrypted data streams from IoT node **IOT-HVF-3906** are compiled into hull dossier **ECO-HVF-2026-V2-663** prior to advancing the assembly to the subsequent bay.

6.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **6.0** on MBT T-90S Bhishma / Arjun Mk-1A (Upgraded Turret), the bay controller shall verify that the IoT telemetry gateway is connected and active. Numquam blanditiis repellat quos iure laudantium et neque dolores fugit tenetur ullam officiis voluptas. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 6.0 below.

TABLE 6.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Cant-Angle Running Torque (15°)	**≤ 410 Nm dynamic continuous**	Wireless Telemetry Torque Flange	3 Full Rotations
6.2	Dynamic Stabilization Drift	**≤ 0.08 mrad RMS during shake**	Dual-Axis Optical Gyroscope	5-Min Vibration Cycle
6.3	Emergency Auxiliary Hand Drive	**≤ 30 N handle crank effort**	Digital Hand-Crank Dynamometer	360° Manual Rotation
6.4	True Azimuth Resolver Linearity	**≤ 0.10 mrad maximum deviation**	Master Optical Theodolite Grid	Every 15° Index
6.5	Turret Lock Pin Shear Test	**Zero deflection under 120 kN lock**	Hydraulic Verification Cylinder	Lock Pin Engagement

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-307	IOT-GW-04 / 204-V2	OP-774 (Biometric)	DGQA-83 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 6.0): Ensure all 5 parameters in Table 6.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

APPENDIX A — VERSION 2 ENGINEERING CHANGE ORDER (ECO) & REVISION LEDGER

This Appendix documents the formal Engineering Change Orders (ECO) incorporated into **Turret Mount System Alignment & Ring Gear Fitting (Version 2)** (HVF/AVNL/SOP-MBT/TI-204-V2) for Version 2.0 at Heavy Vehicles Factory (HVF), Avadi. Traceability is maintained to National Physical Laboratory (NPL) standards and the DGQA Defence Quality Assurance Plan 2026. Temporibus deserunt minus sit repellendus tempore quasi saepe doloribus adipisci necessitatibus iusto dolor.

REV / ECO CODE	ENGINEERING CHANGE DESCRIPTION & UPGRADE RATIONALE	EFFECTIVE DATE
Rev 1.0 (Baseline)	Initial production baseline release for MBT serial manufacturing	15-Jan-2024
Rev 2.0 (ECO-HVF-2026-TUR-419)	Upgraded to Industry 4.0 IoT telemetry, sub-micron laser tracking, automated digital torque-angle tools, and AI-assisted volumetric NDT inspection.	**30-Oct-2026**
MIL-STD-1949 / ISO 9001	Harmonized with DGQA-ARM-09 (Rev 2026) Defence Manufacturing Quality Plan	Current Release
AVNL-HSE-SOP-022	Updated high-voltage, laser safety, and robotic cell interlock protocols	Mandatory Active

APPENDIX B — VERSION 2.0 MULTI-TIER OFFICIAL SIGN-OFF & RELEASE MATRIX

Version 2.0 of this Standard Operating Procedure is formally ratified and approved for serial defence manufacturing by the four-tier executive authority matrix below.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Advanced Mfg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[VER 2.0 APPROVED] DGM/AM/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & NDT) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[VER 2.0 CONCURRED] JGM/QA/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[VER 2.0 COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[RELEASED FOR DEFENCE MFG] CGM-OFFICE/AVNL**	30-Oct-2026

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